F-REGAL: Improving Federated Graph Alignment with Structural Identity and FedSpray Integration

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Abstract

Federated Graph Learning (FGL) faces significant hurdles due to data heterogeneity and the "majority class poisoning" effect inherent in standard Graph Neural Network (GNN) message passing. While existing methods like FedSpray [5] address class-wise biases through feature-structure encoding, they often lack the fine-grained structural awareness required for precise node-to-node alignment across disjoint silos. In this work, we propose F-REGAL, a hybrid framework that integrates REGAL (RElational Graph ALignment) [3] with FedSpray. F-REGAL leverages structural identity-defined via logarithmic degree binning and k-hop neighbor aggregation and uses xNetMF for cross-network matrix factorization within a privacy-preserving federated setting. By combining FedSpray's knowledge distillation correction mechanism with REGAL's sub-quadratic node matching, our approach enables alignment-aware embeddings that handle millions of nodes while maintaining strict data sovereignty. Our empirical results validate the core framework, achieving a 98.0% alignment accuracy utility bound on the PubMed dataset, while quantifying the "privacy tax" introduced by Laplacian Differential Privacy. Furthermore, the aligned representations seamlessly integrate into a robust federated GCN, stabilizing at approximately 86% test accuracy across clients.

1 Introduction

Graph-structured data is ubiquitous in sensitive domains such as financial fraud detection, supply chain management, and medical informatics. However, privacy regulations often prevent the centralization of these graphs, necessitating Federated Learning (FL) approaches. A critical challenge in this setting is Graph Alignment: identifying corresponding nodes across different networks (e.g., matching a user across Twitter and LinkedIn) without sharing raw edge data.

Traditional GNNs in FL suffer from two main flaws: (1) Homophily Assumption, where connected nodes are assumed to share labels, and (2) Message-Passing Bias, where minority nodes are overwhelmed by features from majority-class neighbors in "bad neighborhoods". While FedSpray [5] provides a mechanism to correct these biases using synthetic "structure proxies," it primarily operates at a class-wise level. Conversely, REGAL [3] provides a powerful framework for structural identity-based matching but lacks a native federated integration that accounts for local model drift.

This report presents F-REGAL, which bridges this gap. By utilizing Nyström approximation and xNetMF, we allow clients to compute similarity to global landmarks locally, ensuring that raw edges never leave the client.

2 Related Work

Federated Graph Learning (FGL) Recent works such as FedAvg [4] and APFL attempt to handle data heterogeneity by personalizing local models. However, as shown in previous studies on the PubMed dataset, these methods often struggle with minority node accuracy.

Graph Alignment REGAL [3] introduced the concept of structural identity using k-hop degree descriptors. By projecting these descriptors into a shared latent space via xNetMF, nodes can be matched across networks with sub-quadratic complexity. Our work extends this to a federated context.

3 Method

F-REGAL combines REGAL's structural alignment with FedSpray's federated orchestration to align nodes across private graph silos without sharing raw edges or node attributes. The pipeline is designed

so all clients produce embeddings in the same latent coordinate system, enabling server-side matching.

3.1 Federated Structural Configuration (Server-Side)

Before local computation, the server broadcasts a shared structural configuration:

- global maximum degree limit $d_{max}^{(g)}$ for consistent feature dimensionality.
- neighborhood depth K and discount factor δ for structural identity extraction, shared basis size p (landmark count) and basis version identifier.

This configuration enforces a common feature space of width $K(d_{max}^{(g)} + 1)$ across all clients, preventing index-space and dimensional mismatches during alignment.

3.2 Local Structural Fingerprinting (Client-Side)

Each client k owns a private graph $\mathcal{G}^{(k)} = (V^{(k)}, E^{(k)})$ and computes node structural identity locally:

1. Degree-bucket encoding: node degrees are mapped to logarithmic bins for robustness to local degree variance.
2. K-hop neighborhood aggregation: for each node v_i, structural descriptors are accumulated over expanding neighborhoods with decay δ.
3. Shared-basis projection input: each client forms structural feature vectors with the server-specified dimensionality, then computes local similarity terms required for xNetMF-style projection.

No raw adjacency matrix, edge list, or raw attributes are transmitted.

3.3 Shared Representation via Federated xNetMF

Let $F^{(k)} \in \mathbb{R}^{|V^{(k)}| \times K(d_{max}^{(g)}+1)}$ denote client features. Using the shared basis, clients compute local similarity blocks that are compatible across silos. The server then coordinates a common transform (Nyström/xNetMF style) and obtains aligned embeddings

$$Z^{(k)} \in \mathbb{R}^{|V^{(k)}| \times d}$$

where all $Z^{(k)}$ lie in the same latent space. This removes the centralized requirement to build a single block-diagonal adjacency matrix while preserving structural comparability.

3.4 Server-Side Alignment

The server receives only low-dimensional embeddings and performs node matching using nearest-neighbor retrieval (KD-tree), reusing REGAL's top-n alignment logic. For client pair (a, b), similarity scores are computed from embedding distances and stored in sparse form:

$$S_{ij}^{(a,b)} = exp(-||z_i^{(a)} - z_j^{(b)}||_2).$$

The output is a set of candidate correspondences with confidence scores and metadata (basis version, embedding version, client pair id).

3.5 FedSpray Synergy

FedSpray [5] is used as the federated coordination layer and correction mechanism:

- orchestrates round-based communication and aggregation control.
- supports structurally informed correction through knowledge distillation signals.
- preserves privacy constraints while enabling cross-silo alignment outputs.

In this hybrid design, FedSpray handles federated training logistics, while REGAL provides the node-level alignment engine.

Algorithm 1: F-REGAL Federated Alignment Protocol

Input: Client graphs $\{\mathcal{G}^{(k)}\}_{k=1}^{m}$, shared config $(d_{max}^{(g)}, K, \delta, p)$
Output: Top-n node alignments between selected client pairs
Server broadcasts shared structural config and basis metadata;
for *each client k in parallel* **do**
  Compute local structural features $F^{(k)}$ with width $K(d_{max}^{(g)} + 1)$;
  Compute shared-basis similarity quantities locally;
  Send privacy-safe representation payload to server;
Server computes or applies shared transform to obtain $\{Z^{(k)}\}_{k=1}^{m}$;
for *each requested client pair (a, b)* **do**
  Build KD-tree on $Z^{(b)}$, query top-n neighbors for $Z^{(a)}$;
  Compute sparse similarity scores and return candidate matches;

Table 1: Synergy Analysis: FedSpray vs. F-REGAL

Feature	**FedSpray Role**	**F-REGAL Role**	**Hybrid Advantage**
Primary Function	Federated orchestration and correction	Node-level structural alignment	Privacy-preserving cross-silo matching
Granularity	Class/proxy-informed supervision	Node-specific correspondences	Fine-grained structural consistency
Scalability	Client-round efficient coordination	Sub-quadratic approximate matching	Large-graph alignment under federation

4 Experimental Results

We deployed the complete F-REGAL pipeline using the official PubMed dataset under a realistic cross-silo split, yielding Client A and Client B with substantial topological overlap.

4.1 The Privacy vs. Utility Tradeoff

The most significant finding of our evaluation is the quantification of the "Privacy Tax" associated with structural graph alignment.

- **Maximum Utility Bound (Zero Noise):** When exact structural fingerprints were computed and shared, F-REGAL achieved a 98.0% alignment accuracy, successfully identifying 13,979 out of 14,262 true overlapping nodes. This confirms the mathematical validity of the F-REGAL extraction and projection architecture.
- **Maximum Privacy Bound (Differential Privacy):** To thwart structural triangulation and reconstruction attacks, Laplacian Noise (scale = 0.05) was applied locally to the embeddings prior to server aggregation. Under these privacy constraints, alignment accuracy dropped to 11.9%.

This highlights that while F-REGAL fundamentally solves the federation gap in graph alignment, strict adherence to differential privacy obfuscates structural nuances. The future of federated graph alignment must focus on tuning this parameter to find the "Goldilocks" zone—where representations are blurry enough to ensure privacy, but distinct enough for meaningful GNN evaluation.

4.2 FedSpray GCN Training Convergence

Following secure structural alignment, the feature matrices were passed into a robust Graph Convolutional Network (GCN) built upon the FedSpray architecture, utilizing Dropout and L2 Regularization. Over 30 communication rounds using standard Federated Averaging (FedAvg) [4], local client losses smoothly decayed (e.g., Client A from 1.0140 to 0.3523). Crucially, global blind test accuracy steadily improved, stabilizing at 86.07% for Client A and 85.96% for Client B. This demonstrates that F-REGAL's

structurally aligned outputs are highly compatible with and effective for real-world federated deep learning classification tasks.

5 Conclusion

F-REGAL represents a significant step forward in privacy-preserving graph alignment. By combining the structural identity principles of REGAL [3] with the correction mechanisms of FedSpray [5], we provide a scalable, communication-efficient, and privacy-first solution for cross-client node matching. Our concrete implementation proves the high utility of our structural alignment methodology (98% accuracy) while transparently demonstrating the privacy-utility tradeoff required in real-world deployments. Future work will investigate the robustness of these alignment-aware embeddings under extreme data non-IID conditions, alongside optimizing the differential privacy noise thresholds to maximize GNN performance without compromising patient sovereignty.

References

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